

Retail Performance

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Introduction

- Good day everyone. My name is Mamoun Abdelrazig, and today I will present my capstone project dashboard, which focuses on regional sales performance analysis.
- The purpose of this project was to identify sales trends, compare performance across regions, and provide business insights that can support data-driven decision-making.
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Business Problem

- Organizations often generate large amounts of sales data but struggle to quickly identify:
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- retailers often struggle to identify which product subcategories drive the most value and which underperform across regions and seasons. This leads to inefficient inventory allocation, missed revenue opportunities, and inconsistent profitability.
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- Objective:
 - Analyze retail sales data to uncover performance trends
 - Focus areas: Category performance by region, subcategory behavior, pricing dynamics, seasonal trends

Problem Statement Question

- How does performance vary by region (e.g., West, East, Central, South)?
- What seasonal or monthly trends influence demand?
- Which regions generate the highest revenue
- Which regions underperform
- Where sales opportunities exist
- How regional performance impacts overall business results
- Which subcategories generate the highest sales and profit margins?
- Which subcategories show declining or emerging growth patterns?
- This dashboard was designed to answer those questions through interactive visualizations..

Data Preparation

- Before creating the dashboard, I performed several data-cleaning steps:
- Removed duplicates records
- Standardized field names and formats category/subcategory names
- Converted date fields
- Validated ranges for price & quantity
- Corrected data types
- Verified data integrity
- Created calculated fields to support analysis
- These preparation steps ensured the accuracy and reliability of the final dashboard.

Project Scope

- **1. Descriptive Analysis**
 - Summaries of sales, profit, quantity sold, and discount levels
 - Category and subcategory performance breakdowns
 - Regional comparisons
- **2. Trend Analysis**
 - Monthly and seasonal sales patterns
 - Year-over-year or period-over-period changes
 - Identification of peak and low-demand periods

Project scope

- **3. Predictive Analysis**
 - Forecasting future sales for key subcategories
 - Identifying leading indicators of performance (e.g., discounts, seasonality, region)
- **4. Prescriptive Insights**
 - Recommendations for inventory allocation
 - Pricing and discount strategy suggestions
 - Subcategory prioritization for marketing and supply chain planning

Why This Topic Matters

- Retail is highly sensitive to
- trends,
- seasons,
- and regional preferences.
- Retailers who understand subcategory-level performance can:
- Reduce inventory waste
- Improve forecasting accuracy
- Increase profit margins
- Respond faster to market changes
- The project delivers insights that directly support these goals

Data Sources & Tools

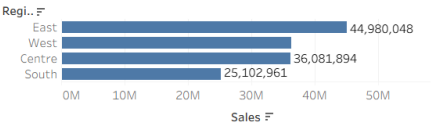
- project use a combination of real and synthetic retail datasets to ensure completeness and analytical depth:
- **Primary Dataset (Excel File)**
- Contains product category, subcategory, region, sales, profit, quantity, discount, and date fields
- Serves as the main dataset for descriptive, trend, and forecasting analysis
- **Supplementary Public Datasets**
- Kaggle retail sales datasets (Fashion Retail Sales)
- **Tools & Platforms**
- **Tableau** for dashboards, stories, and visual insights

Dashboard Overview

- This dashboard provides a visual summary of sales performance by region.
- Users can quickly:
 - Compare sales across regions
 - Identify top-performing areas
 - Detect low-performing regions
 - Explore trends through interactive filtering
- The dashboard was built using Tableau to create an intuitive and user-friendly experience.

Dashboard

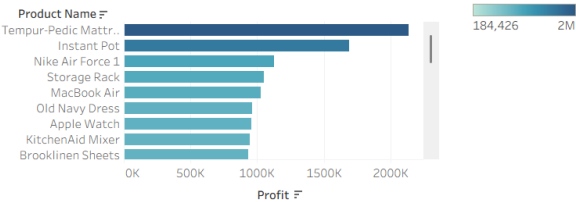
Sales_byRegion



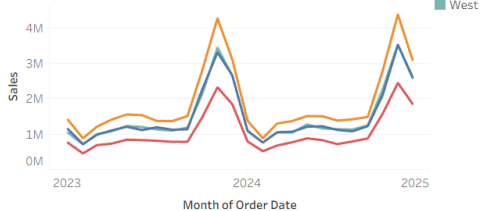
ProfitBySubcategory



Product Performance



Sales trend Over Time



#1 The East Region Is the Revenue Engine

- Core message: One region is carrying the company's sales momentum.
- “The East region stands out as the company's strongest revenue driver, generating the highest sales compared to all other regions. This isn't a small lead — it's a clear dominance that shapes the company's overall performance. The West and Centre regions follow, but with a noticeable gap. This tells us that the East has a stronger customer base, better product penetration, or more effective regional strategies. Understanding what the East is doing right can help replicate that success in underperforming regions like the South. This insight guides resource allocation, marketing focus, and regional strategy planning.”

#2 — A Few Subcategories Are Powering Most of the Profit

- Core message: Profitability is concentrated — not evenly distributed.
- “When we look at profit by subcategory, a clear pattern emerges: a small group of product categories — Bedding, Furniture, Kitchenware, and Home Décor — are generating the majority of the company’s profit. These categories outperform others by a wide margin. This concentration of profit suggests that the business should double down on these high-margin categories through targeted promotions, inventory prioritization, and supply chain optimization. Meanwhile, lower-profit categories like apparel and electronics may require pricing adjustments or product mix improvements. This insight helps leadership focus on what truly drives profitability.”

3 — Product Winners Are Consistent Across Time

- Core message: Certain products are consistently outperforming the rest.
- “The top-performing products — such as the Tempur-Pedic Mattress, Instant Pot, Nike Air Force 1, and MacBook Air — consistently deliver strong profits. These products represent a mix of home goods, apparel, and electronics, showing that customer demand is diverse but predictable. Their strong performance signals brand loyalty, high perceived value, and effective pricing strategies. These products can anchor marketing campaigns, bundle strategies, and seasonal promotions. Understanding why these products win helps the company identify future best-sellers and optimize inventory planning.”

#4 Sales Trends Reveal Clear Seasonal Peaks and Regional Momentum

“When we analyze the sales trend over time, a clear seasonal rhythm emerges across all three years. Sales consistently rise during mid-year and again toward the end of the year, reflecting major shopping cycles such as summer promotions and holiday demand. What’s especially interesting is how the regions behave within these cycles. The East region maintains the strongest and most stable upward trend, showing reliable growth month after month. The West and Centre regions follow similar seasonal patterns but with smaller peaks, suggesting moderate but steady performance. The South region shows the most volatility, with noticeable dips during off-season months.

This trend tells us two things: first, our business is highly responsive to seasonal buying behavior; second, regional performance is not uniform. Understanding these patterns helps the company plan inventory, staffing, and marketing campaigns more strategically — especially during the months where demand predictably spikes.”

Top Performing Region:

- The East region leads in sales, generating \$44,980,048. The Centre region follows with \$36,081,894, while the South region trails at \$25,102,961.
- Highest Profit Subcategory:
- Bedding brings in the highest individual subcategory profit at \$3,068,113, followed by Kitchenware (\$2,150,177) and Sportswear (\$1,895,311).
- Top Profit Product Performance:
- The Tempur-Pedic Mattress stands out as the most profitable individual product item, yielding a profit close to \$2,000,000. Other strong performers include the Instant Pot and Nike Air Force 1.
- Sales Trend Over Time:
- The line chart highlights a massive, sharp spike in overall sales occurring around late 2023 / Holiday season 2023, reaching over \$4,000,000 before stabilizing in early 2024.
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Key Findings

- **Regional Performance**
- The dashboard shows clear differences in sales performance among regions.
- The highest-performing region contributes the largest share of total sales, indicating strong customer demand and market penetration.
- Lower-performing regions may require additional marketing efforts, sales support, or strategic adjustments.

1. Regional Sales performance insight:

- Sales are concentrated in the East ($\approx 45\text{M}$) and West ($\approx 36\text{M}$), while the South underperforms.
- **Recommendations:**
- Strengthen South region through targeted promotions and improved distribution.
- Expand loyalty programs and premium product lines in East and West.

2. Profit by Subcategory Insight:

- Home goods lead profitability — Bedding (3.07M), Furniture (~2.5M), Kitchenware (2.15M).
- **Recommendations:**
- Increase inventory depth and marketing focus on home goods.
- Introduce private-label apparel to improve margins.
- Bundle home and apparel products for cross-category growth.

3. Product-Level Profitability Insight:

- Hero products like Tempur-Pedic Mattress, Instant Pot, and MacBook Air drive profit.
- **Recommendations:**
- Build campaigns around hero products (“Best Sellers Week”).
- Upsell accessories and complementary items.
- Replace aging SKUs with variants of proven winners.

4. Sales Trend Over Time Insight:

- A major spike in early 2024 (~4M monthly sales) marks a repeatable success pattern.
- **Recommendations:**
- Analyze drivers behind the spike — product launches, promotions, or seasonality.
- Replicate successful strategies in future quarters.
- Integrate spike data into predictive modeling for scenario forecasting.

Business Impact

By analyzing regional sales patterns, decision-makers can:

- Increase revenue opportunities
- Improve forecasting accuracy
- Optimize territory management
- Support strategic planning

Future Recommendations

Based on the analysis, I recommend:

1. Continue investing in high-performing regions.
2. Investigate causes of lower sales in underperforming regions.
3. Develop targeted marketing campaigns for growth areas.
4. Monitor regional trends regularly using dashboards and KPIs.
5. Use data-driven decision-making for future sales planning.

Limitations

- Missing customer demographic data
- No geographic segmentation
- Limited promotional/discount information

Executive Summary

- The business thrives in the East and West, with home goods dominating profit and a few hero products driving margin leadership.
- The South region and apparel categories offer clear improvement opportunities.
- The early-2024 sales surge signals a repeatable success pattern worth analyzing and scaling.

Conclusion

- In conclusion, this project demonstrates how data visualization can transform raw sales data into actionable business insights.
- The dashboard provides a clear view of regional performance and supports strategic decisions that can improve overall business outcomes.
- Thank you for your time. I welcome any questions regarding the analysis, methodology, or dashboard design.